FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Main)-2021

TEST DATE: 18-04-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

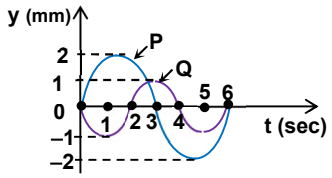
Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

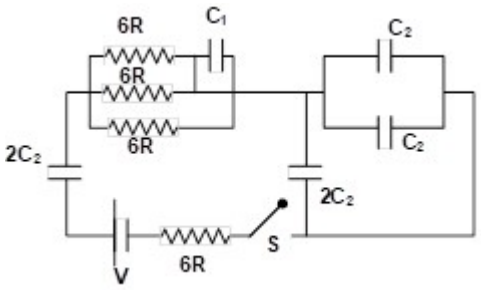
1. The figure shows the displacement time graph for a particle due to two infinite sinusoidal wave forms P and Q, which travel along a stretched string when it is separately excited at two different frequencies. The string is always kept under a fixed tension. The ratio of power transmission by waves P and Q is



(A) 1
(B) 4/3
(C) 16/9
(D) none of these

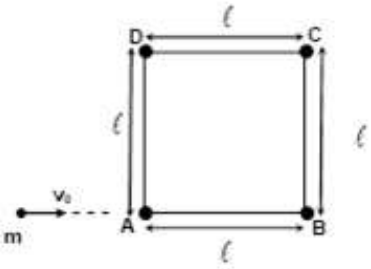
2. A particle of mass m is released on a smooth inclined plane having inclination θ.Then choose correct option.
(A) vertical component of acceleration is g.
(B) vertical component of acceleration of particle is g sin θ.
(C) horizontal component of acceleration of particle is g sin θ cos θ.
(D) horizontal component of acceleration of particle is g cos 2θ.

3. For the circuit shown in figure, which statement is true about the time constant of charging if switch is closed. It is given $C_1 = C$, $C_2 = C$



(A) time constant for charging of C_1 is 3 RC
(B) time constant for circuit is $\frac{3RC}{5}$
(C) time constant for circuit is $\frac{32RC}{3}$
(D) none of these

4. Four particles (A, B, C and D) each of mass m are connected by four massless rods each of length ℓ. All four particles lie on a smooth horizontal plane. A particle of mass m moving along the rod AB with a velocity v_0 strikes the particle A and stops (as shown in figure) Find the angular velocity of each particle after collision.



(A) $\frac{v_0}{\sqrt{3}\ell}$
(B) $\frac{v_0}{4\ell}$
(C) $\frac{v_0}{\ell}$
(D) $\frac{v_0}{2\ell}$

5. A particle of mass m_1 moving with a velocity v_0 and collides head on with stationary particle of mass m_2. After the collision, the velocities of the particles are v_1 and v_2. The condition for v_1 to be positive i.e. for the first particle to continue moving in the same direction is (e = coefficient of restitution)

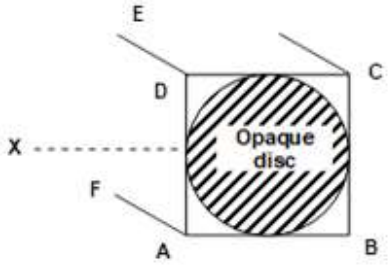
(A) $\frac{m_1}{m_2} > 1$

(B) $\frac{m_1}{m_2} < 1$

(C) $\frac{m_1}{m_2} > e$

(D) $\frac{m_1}{m_2} < e$

6. A solid cube of edge length 2m and index of refraction 5/3 has a point source at its center. An opaque disc of radius 1 m covers the face ABCD as shown in figure. The radius of disc starts contracting for $0 \le t \le 4$ sec and given as $r = (1 - kt)$ m, where $k = \frac{1}{8}\text{sec}^{-1}$. The time when the source is just visible from the face ABCD is



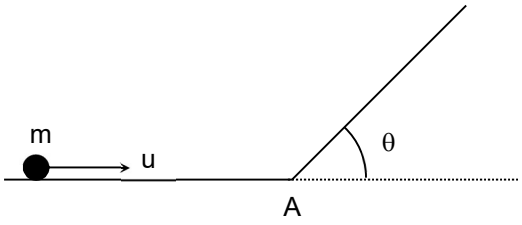
(A) 2 sec

(B) $\frac{4}{3}$sec

(C) $\frac{8}{5}$sec

(D) none of these

7. In the figure, a small ball is moving with a velocity u on a frictionless horizontal surface. At point A, it encounters an inclined plane and collides with it elastically. What will be the vertical component of velocity of ball just after collision (all surfaces are frictionless)



(A) u cos θ sin θ

(B) u sin 2θ

(C) u(sin θ + cos θ)

(D) u sin θ

8. A body is moved along a straight line by a machine delivering power proportional to time t. The distance moved by the body in time t is proportional to

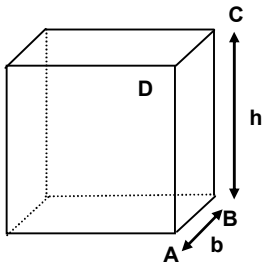
(A) $t^{1/2}$

(B) $t^{3/4}$

(C) $t^{3/2}$

(D) t^2

9. A liquid of variable density $\rho = \rho_0\left(1+\frac{x}{h}\right)$ is filled in a cuboidal vessel up to height h, where x is the distance from top. Find the force applied by the liquid on the vertical face ABCD. (ρ_0 is constant and neglect atmospheric pressure)



(A) $\frac{4}{3}\rho_0 bgh^2$

(B) $\frac{2}{3}\rho_0 bgh^2$

(C) $\frac{1}{2}\rho_0 gbgh^2$

(D) $\frac{5}{6}\rho_0 bgh^2$

10. A particle starts (from rest) moving on a circular path of radius R with uniform angular acceleration α. The magnitude of average linear velocity during time interval t = 0 to t = t is

(A) $\frac{R}{t}\sin\alpha t^2$

(B) $\frac{2R}{t}\sin\frac{\alpha t^2}{4}$

(C) $\frac{2R}{t}\sin\frac{\alpha t^2}{2}$

(D) $\frac{R}{t}\sin\frac{\alpha t^2}{4}$

11. A particle moving horizontally with a velocity u strikes a fixed frictionless sphere at height R/4 above the center of sphere (collision is completely inelastic). The velocity of particle just after collision is

(A) $\frac{u}{4}$

(B) $\frac{u}{5}$

(C) $\sqrt{2}u$

(D) $\frac{\sqrt{2}u}{5}$

12. One mole of an ideal monoatomic gas is taken from temperature T_0 to $3T_0$ by the process PT^{-5} = constant. Then choose the correct alternative.

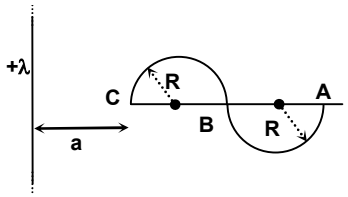
(A) molar heat capacity of gas is $\frac{3R}{2}$

(B) molar heat capacity of gas is $-\frac{5R}{2}$

(C) molar heat capacity of gas is $-\frac{3R}{4}$

(D) molar heat capacity of gas is $\frac{5R}{4}$

13. In figure there is an infinitely long uniformly charged non-conducting wire having linear charge density $+\lambda$. Find the net work done in taking a point charge q from A to C through semicircular paths AB and BC.



(A) $\frac{q\lambda}{\varepsilon_0}\ln\left(\frac{a+2R}{a}\right)$

(B) $\frac{q\lambda}{4\varepsilon_0}\ln\left(\frac{a+4R}{a}\right)$

(C) $\frac{q\lambda}{\pi\varepsilon_0}\ln\left(\frac{a+2R}{a}\right)$

(D) $\frac{q\lambda}{2\pi\varepsilon_0}\ln\left(\frac{a+4R}{a}\right)$

14. A wire of mass 200 gm carrying a current of 4A towards increasing x is in the form of $y = 2x^2$ $(-4m \le x \le +4m)$. This wire is placed in a magnetic field $\vec{B} = -0.04\,\hat{k}$ Tesla. The acceleration of wire (in m/s^2) is

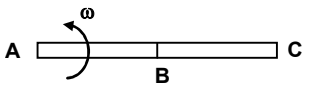
(A) $6.4\,\hat{j}$

(B) $-6.4\,\hat{j}$

(C) $1.28\,\hat{j}$

(D) $12.8\,\hat{j}$

15. In figure there is a non-conducting rod AC in which half part AB has no charge and part BC has linear charge density λ. If the rod rotates about end A with a constant angular velocity ω in the plane of paper, then magnetic field at point A will be :



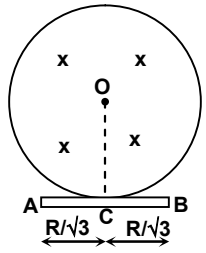
(A) $\frac{\mu_0\omega\lambda}{4\pi}\ln 2$

(B) $\frac{\mu_0\omega\lambda}{\pi}\ln 3$

(C) $\frac{\mu_0\omega\lambda}{4\pi}\ln 6$

(D) $\frac{\mu_0\omega\lambda}{6\pi}\ln 2$

16. A uniform but time varying magnetic field is present in a circular region of radius R. The magnetic field is perpendicular and into the plane of the loop and the magnitude of field is increasing at a constant rate α. There is a straight conducting rod of length $\frac{2R}{\sqrt{3}}$ placed as shown in figure. The magnitude of induced emf across the rod is



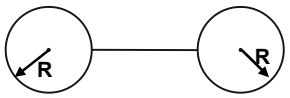
(A) $\frac{\pi R^2\alpha}{4}$

(B) $\pi R^2\alpha$

(C) $\frac{\pi R^2\alpha}{3}$

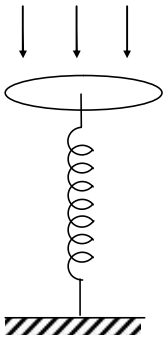
(D) $\frac{\pi R^2\alpha}{6}$

17. In the figure shown two identical conducting spheres have charges 2 μC, radius R = 10 cm each are connected through a conducting rod of length 1 m. The system is now kept in a uniform electric field E = 5 N/C, what will be the electric field at the center of right conducting sphere.



(A) 2 N/C
(B) 3 N/C
(C) 7 N/C
(D) zero

18. In the figure shown, radiation of intensity I is falling on a disc of mass m and radius R, attached to a spring of spring constant K. If 20% of radiation is reflected, what will be the compression in the spring in equilibrium? (C is the speed of light)



(A) $\frac{1}{K}\left[mg+\frac{2I\pi R^2}{C}\right]$

(B) $\frac{1}{K}\left[mg+\frac{1.2I\pi R^2}{C}\right]$

(C) $\frac{1}{K}\left[mg+\frac{I}{C}\right]$

(D) $\frac{1}{K}\left[mg+\frac{I\pi R^2}{C}\right]$

19. A stationary hydrogen atom emits photon corresponding to the first line of Balmer series. If R is the Rydberg's constant and M is the mass of atom, then the velocity acquired by atom is

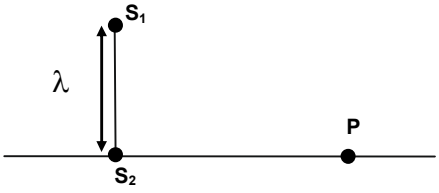
(A) $\frac{5}{36}\frac{hR}{M}$

(B) $\frac{3}{17}\frac{hR}{M}$

(C) $\frac{3}{4}\frac{hR}{M}$

(D) $\frac{5}{4}\frac{hR}{M}$

20. Two coherent point sources S_1 and S_2 emit monochromatic light of wavelength λ. The separation between the sources is λ. Consider a line passing through S_2 and perpendicular to line S_1S_2. Find the distance from S_2 on line S_2P, where the intensity is minimum.



(A) $\frac{5\lambda}{12}$

(B) $\frac{7\lambda}{12}$

(C) $\frac{3\lambda}{4}$

(D) $\frac{5\lambda}{8}$

SECTION – B
(Numerical Answer Type)

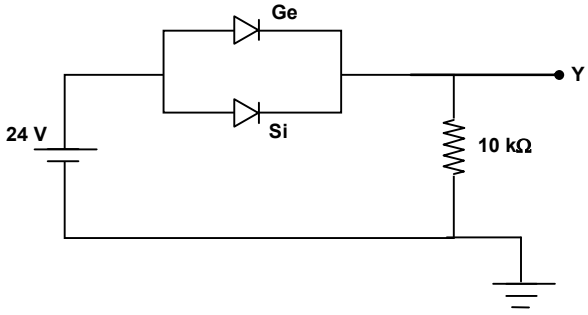
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. Average life of a radioactive sample is 4 ms. Initially the total number of nuclei is N_0. A charged capacitor of capacity 40 μF is connected across a resistor R. What will be the value of R (in Ω) if the ratio of number of nuclei remaining to charge on the capacitor remains constant with time.

22. In series circuit consisting of a capacitor with capacitance C = 200 μF and a coil with active resistance R = 20 Ω and inductance L = 0.35 H is connected to a source of alternating voltage with amplitude V_m = 180 volt and angular frequency ω = 100 sec^{-1}. Find the amplitude of voltage across the capacitor in volt.

23. A siren creates a sound level of 60 dB at a location of 500 m from speaker. The siren is powered by a battery that delivers a total energy of 1 kJ. Assuming the efficiency given to be 20%, determine the total time (in sec) for which the siren sounds. (Take π = 3.14)

24. Two junction diodes one of Germanium (Ge) and other of Silicon (Si) are connected as shown in figure to a battery of emf 24 volt and a load resistance 10 kΩ. The germanium diode conducts at 0.3 volt and silicon at 0.7 volt. When a current just starts flowing in the circuit, what will be the potential of terminal Y (in volt) just after that instant.



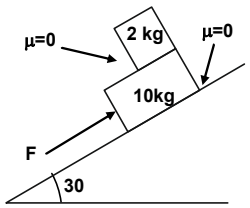
25. Two Polaroid's are oriented with their planes perpendicular to incident light and transmission axes making an angle 60° with each other. What percentage of intensity of incident unpolarised light is transmitted?

26. Earth's magnetic field may be supposed to be due to a small bar magnet located at the center of the earth. If the magnetic field at a point on the magnetic equator is 0.6 × 10^{-4} Tesla. The magnetic moment of such a bar magnet is N × 10^{22} Am^2. Find N. (Radius of earth = 6400 km)

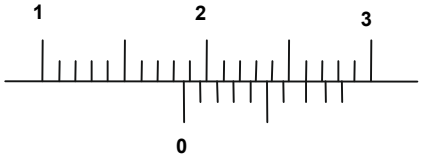
27. What will be the time period (in second) of a simple pendulum having length equal to four times the radius of earth. (Given radius of earth = 6400 Km, g = 10 m/s^2, π = 3.14 and $\sqrt{5} = 2.24$)

28. A person brings a mass 2 kg from infinity to a point A. Initially, the mass was at rest but it is moving at a speed of 4 m/sec when it reaches point A. The work done by the person on the mass is -10.5 Joule. What is the magnitude of the gravitational potential at A in Joule/kg.

29. In the figure shown, what can be the value of F in newtons for which 10 kg block will be in equilibrium? (All the surfaces are frictionless and g =10 m/s^2)



30 There is a vernier calliper having 1cm divided into 10 equal divisions on the main scale. The vernier scale has 10 equal divisions that correspond to 11 main divisions. The reading of the calliper is shown in figure. The measured value (in cm) by the vernier calliper is



Chemistry

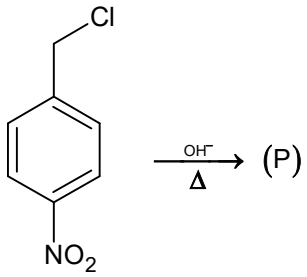
PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

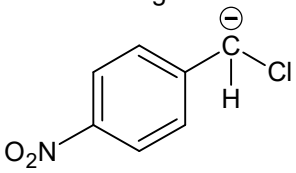
31. The magnetic moment of two ions M^{x+} and M^{y+} of the element M(Z = 26) is found to be 5.916 BM. If x > y, then which of the following statement is correct?
(A) M^{y+} is more stable than M^{x+}
(B) M^{y+} is less stable than M^{x+}
(C) Both are equally stable
(D) Can't be predicted

32. Which of the following cases hybridisation of underlined element is changing?
(I) $\underline{Al}(OH)_3$ precipitate dissolved in NaOH
(II) $\underline{B}_2H_6$ is dissolved in THF
(III) $\underline{Si}F_4$ vapour is passed through liquid HF
(IV) Solidification of $\underline{P}Cl_5$ vapour.
(A) I and IV
(B) I, II and IV
(C) I, III and IV
(D) II, III and IV

33. 

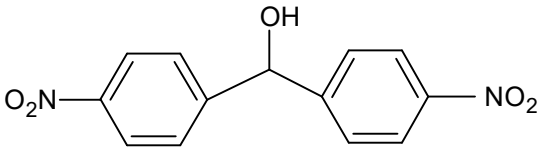
(A)

Which of the following is not correct about above reaction?

(A)  is the intermediate.

(B) 4, 4′ – dinitro Stilbene is final product (P)

(C) Elimination is also taking place to get final product.

(D)  forms when intermediates combines with A.

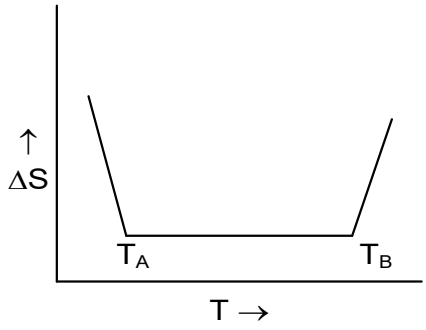
34. One litre of an unknown gas weighs 1.25 g at NTP. Which of the following gas pertains to the above statement?
(A) CO_2
(B) NO_2
(C) N_2
(D) O_2

35. Which of the following is correct?
(A) Specific conductivity of a solution decreases with dilution whereas molar conductivity increases with dilution.
(B) Specific conductivity of a solution increases with dilution, whereas molar conductivity decreases with dilution
(C) Both specific conductivity and molar conductivity decreases with dilution.
(D) Both specific conductivity and molar conductivity increases with dilution

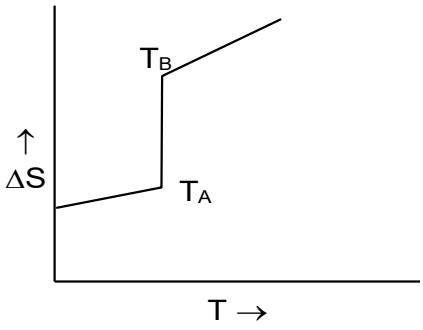
36. If the total vapour pressure of a liquid mixture A and B is given by the equation $P = 150\ x_A + 50$ then the ratio of the vapour pressure of pure liquid A and B is given by
(A) 3 : 2
(B) 4 : 1
(C) 3 : 1
(D) 6 : 2

37. T_B and T_A are melting point and freezing point of a substance X. Which of the following is correct variation of entropy by graph between entropy change (ΔS) and temperature is

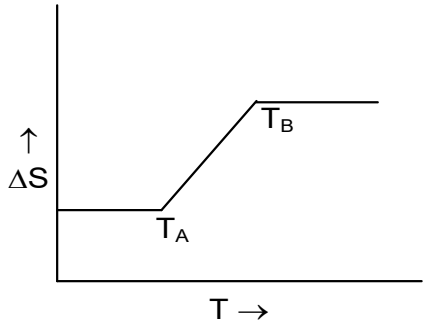
(A)



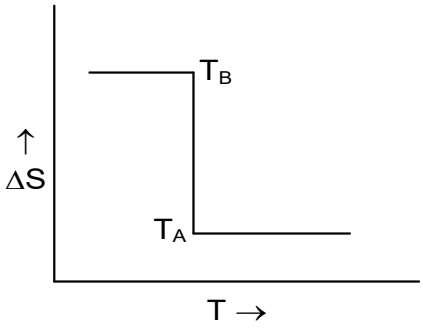
(B)



(C)

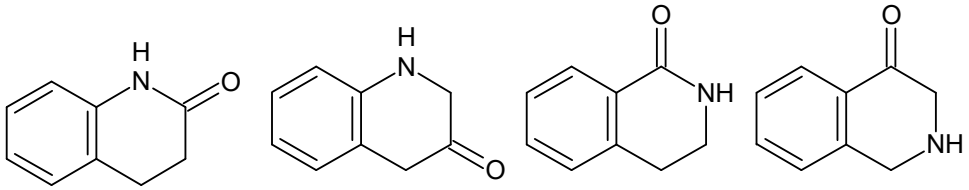


(D)



38. If a freshly formed precipitate of SnO_2 is peptised by a small amount of NaOH, these colloidal particles may be represented as
(A) $[SnO_2]Na^+ : OH^-$
(B) $[SnO_2]Sn^{+4} : OH^-$
(C) $[SnO_2]\,SnO_3^{-2} : 2Na^+$
(D) $[SnO_2]SnO^{-2} : Na^+$

39. Arrange the following compounds in decreasing order of rate of electrophilic aromatic chlorination with $Cl_2/FeCl_3$

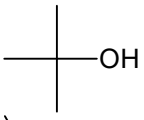


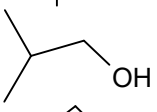
(A) (B) (C) (D)

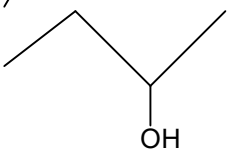
(A) A > B > D > C
(B) B > A > C > D
(C) B > A > D > C
(D) B > C > A > D

40. Arrange the following alcohols in decreasing boiling pont

(A)


(B)


(C)


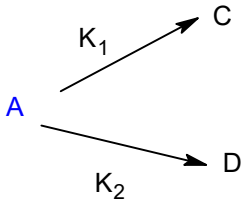
(D)


(A) A > D > C > B
(B) B > D > A > C
(C) D > B > C > A
(D) A > C > D > B

41. Which of the following statement is correct?
(A) Alkynes are more reactive than alkenes towards their reaction with HCl
(B) Beryllium carbide on hydrolysis produces methane
(C) Cyclohexene gives different products with HBr in the presence and absence of benzoyl peroxide
(D) 2-fluorobutane on reacting with alcoholic KOH gives but-2-ene as a major product

42. In a nitrosyl complex the odd electron on nitrogen of the coordinated nitrosyl is transferred to vacant (n–1)d sub shell of the central metal. Hence the spin only magnetic moment of the complex $K[Cr(CN)_4(NH_3)(NO)]$ assuming the metal ion to be in the strong field is:
(A) $\sqrt{8}$ BM
(B) O BM
(C) $\sqrt{24}$ BM
(D) $\sqrt{15}$ BM

43. Consider the following case of competing 1st order reaction:



After the start of the reaction at t = 0 with only A, the [C] is equal to the [D] at all times. The time at which all three concentrations will be equal is given by

(a) $t = \frac{1}{K_1 + K_2}\ln 3$ (b) $t = \frac{1}{2K_2}\ln 3$

(c) $t = \frac{1}{2K_1}\ln 2$ (d) $t = \frac{1}{3K_2}\ln 2$

(A) Only (a)
(B) (a) and (b)
(C) (a), (b) and (c)
(D) None of these

44. An inorganic salt in its aqueous solution produced a white ppt with NaOH which dissolves in excess of NaOH. Also its aqueous solution produced light yellow ppt. with $AgNO_3$ sparingly soluble in NH_4OH. The probable salt is
(A) $AlBr_3$
(B) AlI_3
(C) $AlCl_3$
(D) $ZnCl_2$

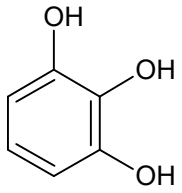
45. Which one is not carbonate ore
(A) Cerussite
(B) Crocoite
(C) Siderite
(D) Calamine

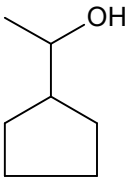
46. Which statement is incorrect?
(A) I_2, KI, rectified spirit and H_2O is Tincture of iodine
(B) Among halogens I_2 is the most reactive with HNO_3
(C) Cl_2 is used as cutting wool
(D) Cl_2 gas can be dried over concentrated sulphuric acid

47. Which of the following is (are) alloy(s) not containing copper?
(I) Bell metal
(II) Brass
(III) Stainless steel
(IV) Solder
(A) Only (I)
(B) (II) and (III)
(C) (III) and (IV)
(D) (I), (III) and (IV)

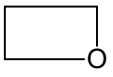
48. Which of the following is correct?
(A) B < Al < Ga < In < Tl (Atomic size)
(B) $Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$ (Solubility)
(C) LiOH < NaOH > KOH > RbOH < CsOH (solubility)
(D) $BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$ (Solubility)

49. Which of the following is not correctly named

(A) Pyrogallol 

(B) Cyclopentylmethylcarbinol 

(C) Acrolein $CH_2 = CH - CN$

(D) Oxetane 

50. Which one of the following compounds is not a vitamin?

(A) Ascorbic Acid
(B) Thiamine
(C) Testosterone
(D) Riboflavin

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. In the reaction with sodium hydroxide, how many of the following substances are oxidised?
Be, B, C, Al, Si, P, S, Sn, Pb, NH_4^+, $AlCl_3$, $FeCl_3$, $SnCl_2$.

52. How many geometrical isomers can be shown by 1, 2, 3-trichlorocyclohexane?

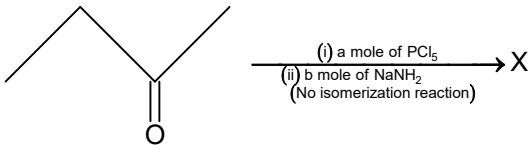
53. 50 ml 0.1 M NH_4OH is mixed with 50 ml 0.05 M HCl. Find the pH of the resultant mixture. $\left[pK_b = 4.7447\right]$

54. In Duma's method for estimation of nitrogen 0.125 g of an organic compound gave 20 ml of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is:

55. 24×10^{23} protons are removed from 4 grams of a lightest gas known. Then the charge left over is 'X' Faraday. Calculate X [Given : Avagadro's Number = 6×10^{23})

56. What will be the percentage of free volume available in 1 mole of $H_2O(g)$ at 760 mm and 373 K? Density of H_2O (l) at 373 k is 0.953 g/ml

57. A copolymer of ethene and vinyl chloride contains alternate monomers of each type. What is the mass percentage of vinyl chloride in this copolymer?

58. 

Where X is molar mass of organic product.

Find the value of $\left(\frac{X}{a+b}\right) \div$ Mass of H in the product . (At. Mass of H = 1 amu)

59. One gram mixture of CH_3OH and CH_3CHO reacts with Benedict's reagent to give a red precipitate. The mass of the red precipitate obtained is 2.86 g. Calculate the % of CH_3CHO in the mixture.

60. A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms (A) and (B). The form (A) reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, where as (B) gives a pale yellow precipitate soluble in concentrated ammonia. Magnetic moment of complexes A or B is (in Bohr magneton)

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The sum of the series $\frac{13}{1.2.3.2}+\frac{26}{2.3.4.4}+\frac{43}{3.4.5.8}+\frac{64}{4.5.6.16}+.....\infty$

(A) $\frac{3}{4}$

(B) $\frac{7}{4}$

(C) $\frac{3}{2}$

(D) $\frac{4}{3}$

62. Let N be the number of ways in which 20 persons can be made to seat around four round tables painted with distinct colours such that there are 5 persons around each table, then numbers of zeros at the end of N is/are

(A) 0
(B) 1
(C) 3
(D) 4

63. A natural number is selected from 1 to 1000 at random, then the probability that a particular non-zero digit appears at most once is

(A) $\frac{3}{250}$

(B) $\frac{143}{250}$

(C) $\frac{243}{250}$

(D) $\frac{7}{250}$

64. If $\sum_{r=0}^{n}\frac{(-2)^r\ {}^nC_r}{{}^{r+2}C_r}=\frac{1}{101}$, then the value of n is

(A) 97
(B) 98
(C) 99
(D) 101

65. If P(Z), Q(Z^2), R(Z^3) are the vertices of a right angled triangle where $\angle PRQ = 90°$, then the locus of Z is

(A) a line, Re(Z) = –1, $Z \neq -1$

(B) a line Re(Z) = 0, $Z \neq 0$

(C) a circle with centre (–1) and radius 1, $Z \neq 0$

(D) a circle with centre $\left(-\frac{1}{2}\right)$ and radius $\frac{1}{2}$, $Z \neq -1, 0$

66. Let $I_1 = \int_0^{\pi/3} (2\sec x)^{100}\, dx$, $I_2 = \int_0^{\ln(2+\sqrt{3})} \left(e^x + e^{-x}\right)^{99} dx$, then $\frac{I_1}{I_2}$ is

(A) 1

(B) 2

(C) $\frac{1}{2}$

(D) 2^{99}

67. Let $f:\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \left(-e^{-\pi/2}, e^{\pi/2}\right)$ be a function define as $f(x) = e^x(\sin x + \cos x)$ and $g(x) = f^{-1}(x)$, then the value of $g''(1)$ is

(A) $-\frac{1}{2}$

(B) $-\frac{1}{4}$

(C) $-\frac{1}{8}$

(D) $\frac{1}{2}$

68. $\int_1^5 \left(\sqrt{x + 2\sqrt{x-1}} + \sqrt{x - 2\sqrt{x-1}}\right) dx$ is equal to

(A) $\frac{8}{3}$

(B) $\frac{32}{3}$

(C) $\frac{34}{3}$

(D) $\frac{64}{3}$

69. If normal at point P(6, 2) on the curve $xy - x - 2y - 2 = 0$ intersects the curve again at $Q(\alpha, \beta)$, then the value of $12\alpha + \beta$ is

(A) 4

(B) 5

(C) 6

(D) 7

70. Let $a = \sin^{-1}\sin(-16)$, $b = \cos^{-1}\cos 16$, then the minimum value of the expression $\frac{5}{2}\cos^2((a+b)x) - \sin(3(a+b)x)$ where $x \in [0, 1]$ is

(A) $\frac{5}{2}$

(B) $\frac{17}{32}$

(C) 1

(D) $\frac{15}{32}$

71. $\lim\limits_{x\to\infty}\left(x^2 + \frac{1}{x}\right)e^{\frac{1}{x}} - x^2$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) $\frac{1}{8}$

(D) 1

72. In ΔABC if $\frac{ar_1 + br_2 + cr_3}{\Delta} = 6$, ($r_1$, r_2, r_3 are the ex-radii), then the triangle is

(A) right angled

(B) right angled isosceles

(C) equilateral triangle

(D) $\angle A = 120°$

73. Let $A \equiv \{x : \sin x + \sin 2x + \sin 3x + \sin 4x = 0 ; x \in [0, \pi]\}$, $B = \{x : \sin 2x > \cos x\}$. Number of elements in $A \cap B$ is/are

(A) 0

(B) 2

(C) 3

(D) 4

74. The logical statement $(p \to \sim q) \vee (q \to p)$ is a

(A) tautology

(B) contradiction

(C) contingency

(D) equivalent to $p \vee q$

75. The variance of first 100 odd natural number is k, then the variance of twice of the same numbers will be

(A) k

(B) 2k

(C) 4k

(D) $\frac{k}{2}$

76. The plane $4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $5x + 3y + 10z = 25$. The equation of plane in its new position is $x - 4y + 6z = k$, where k is
(A) 106
(B) –89
(C) 73
(D) 37

77. If f(x) is a differentiable function satisfying $\int_0^x f(t)dt = 2x^4 - 3x^3 + 6x^2$; f(1) = 6, then the value of f(2) is
(A) 6
(B) 12
(C) 18
(D) 24

78. Let f be a polynomial of degree 5 such that $\lim_{x\to 0}\left(\frac{f(-x)}{2x^4}\right)^{\frac{1}{x}} = \frac{1}{e^4}$, then the value of f(1) is
(A) 10
(B) 11
(C) 12
(D) 15

79. The area bounded by the curve $5\{y\} = 3[x] - 2$; $1 \le y \le 2$ (where {.} ad [.] represents the fractional part and greatest integer function respectively) and x-axis is
(A) 2
(B) $\frac{5}{2}$
(C) $\frac{8}{3}$
(D) 3

80. Let $P(\alpha, \beta)$ be a point on the line $4x + 3y + 21 = 0$. If A(4, 0) and B(0, 4) be two points such that |PA – PB| is minimum, then
(A) $\alpha - 5\beta - 10 = 0$
(B) $\alpha - 5\beta - 12 = 0$
(C) $5\alpha + \beta - 12 = 0$
(D) $\alpha - 5\beta - 15 = 0$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. Let $A = \left[a_{ij}\right]_{2\times 2}$ be a symmetric matrix where $a_{ij} \in \{0, 1, 2, 3\}$. The number of invertible matrices possible are

82. Let $f(x) = ax^2 + bx + c$, $a \neq 0$, a, b, c are integers. If f(1) = 0, 102 < f(6) < 110, 196 < f(8) < 206, then the value of f(10) is

83. The sum of all the real roots of the equation $\left(x^2-8x+6\right)^{x^2-9}=1$ is/are

84. The upper four-fifth portion of a vertical tower subtends an angle $\alpha=\tan^{-1}\frac{2}{3}$ at a point A in the horizontal plane through its foot and at a distance 100 m from the foot. If the angle subtended by the lower one-fifth of tower at point A is β; $\left(\alpha+\beta>\frac{\pi}{4}\right)$, then the height of the tower is

85. Let ΔXYZ be a triangle. Let $\vec{a}=\overrightarrow{YZ}$, $\vec{b}=\overrightarrow{ZX}$, $\vec{c}=\overrightarrow{XY}$. If $|\vec{a}|=6$, $|\vec{c}|=14$, $\vec{a}\cdot\vec{b}=30$, then the value of $\frac{\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}\right|^2}{100}$ is

86. Let S = 0 be the equation of auxiliary circle of the curve $(3x + 4y – 7)^2 + 4(4x – 3y – 1)^2 = 100$. If P be the point on the curve $y^2 + 22y – 2x + 129 = 0$ which is at a minimum distance from the centre C of the circle S = 0, then area of triangle of OCP is (O being origin)

87. Let P and Q be distinct points on the parabola $y^2 = 8x$ such that a circle drawn on PQ as diameter touches the line x + 2 = 0. If length of chord PQ is $\frac{200}{9}$ and the area of triangle PQR, (R is point of intersection of normals at P and Q on the parabola) is k sq. unit, then the value of 27k is

88. The product of the real roots of the equation $\tan^{-1}\cot\left(\frac{3x^2+9|x|+21}{x^2+3|x|+8}\right)=\frac{\pi}{2}+\sec\sec^{-1}\left(\frac{3-27|x|}{9|x|}\right)$

89. Let a, λ, $\mu \in R$, consider the system of linear equations $ax + 3y = \lambda$, $4x – 3y = \mu$;
$\lambda, \mu \in \{–5, –4, –3, –2, –1, 0, 1, 2, 3, 4, 5\}$. If the system of equations has no solution, then the number of ordered triplets (a, λ, μ) possible is/are

90. If ab = 20! ; a, b $\in$ N, then number of ordered pairs of (a, b) such that $\frac{a}{b}$ is an integer are